

1) Downloading the Kubespray Offline Bundle (External Network Environment)

Download all files required for the Kubespray offline installation on a server with internet access running RHEL/Rocky 9 (hereinafter referred to as the "Preparation Server").

1.1. Preparation Server Initial Setup

Install essential tools such as `git` (for source code cloning) and `podman` (for container image management) on the preparation server.

```
# Update system packages to the latest version.
$ sudo dnf update

# Install Git, Podman (or nerdctl), and the Vim editor.
$ sudo dnf install git podman vim
```

1.2. Downloading the Kubespray Offline Bundler

Clone the official `kubespray-offline` repository, which contains the scripts and configuration files required for offline installation.

```
$ git clone <https://github.com/kubespray-offline/kubespray-offline.git>
```

1.3. Configuring Download Versions

Navigate to the cloned directory and configure the versions for Kubernetes and key components (CNI, etcd, etc.) to be downloaded.

```
$ cd kubespray-offline
# Open config.sh to modify KUBE_VERSION, CNI_VERSION, CILIUM_VERSION
```

```
N, etc. (Optional)
$ vim config.sh
```

(image-20251110-091025.png)

1.4. Downloading All Offline Assets

Start downloading all required assets based on the configuration file (`config.sh`).

```
# This script downloads RPM packages, binaries, and all container images,
# saving them into the 'outputs' directory.
$ ./download-all.sh
```

1.5. Manually Adding the Operator Image

To resolve known offline installation bugs in Kubespray v2.29.0 (Issues \#12534, \#12558), you must manually add the `cilium/operator-generic` image required by Kubespray to the bundle, instead of the `cilium/operator` image downloaded by the default script.

```
# Navigate to the 'outputs/images' directory where all container image tarballs are stored.
$ cd outputs/images
```

Preparing the Cilium Operator Generic Image

```
# 1. (Pull) Download the 'operator-generic' image that Kubespray actually intends to run.
$ podman pull quay.io/cilium/operator-generic:v1.18.2

# 2. (Save) Save the downloaded image as a .tar.gz file, following the Kubespray offline naming convention.
$ podman save -o quay.io_cilium_operator-generic-v1.18.2.tar.gz quay.io/cilium/operator-generic:v1.18.2
```

Updating the Image List (`images.list`)

Register the new image in `images.list` so the offline installation script recognizes it and can push it to the private registry.

```
# 3. Open the image list file.  
$ vim images.list
```

Add the source name of the image you just saved as a new line in the file.

```
... (Existing image list) ...  
quay.io/cilium/operator-generic:v1.18.2
```

1.6. Compressing and Moving the Offline Bundle

Compress the `outputs` folder containing all prepared assets to get ready for transfer to the air-gapped network.

```
# Navigate to the parent directory (kubespray-offline).  
$ cd ../../  
  
# Compress the entire 'outputs' folder into a .tar.gz file.  
$ tar -zcvf outputs.tar.gz outputs  
  
# Copy the generated 'outputs.tar.gz' file to the Bastion server  
# in the air-gapped environment via USB or a secure channel.
```